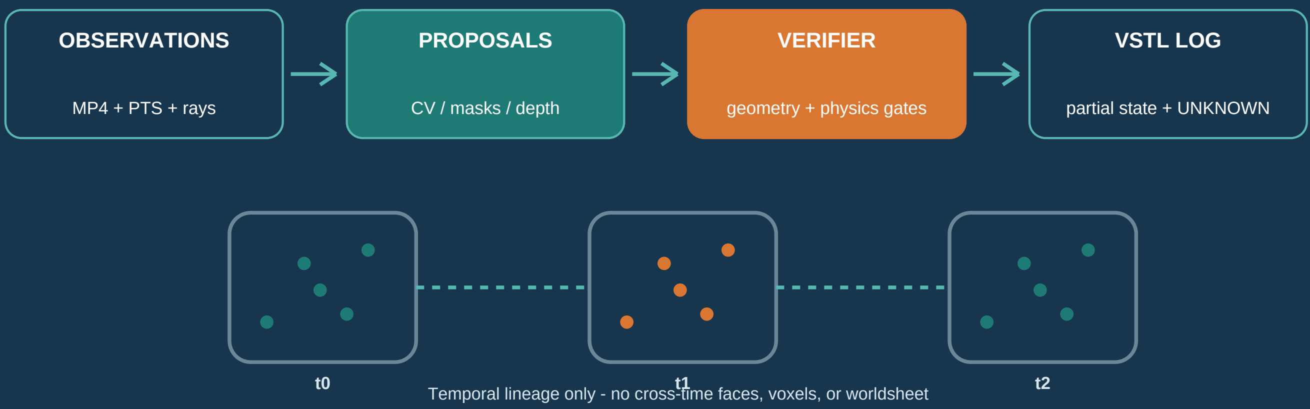
UGTS / VISION SPATIAL-TEMPORAL LITERAL

Monocular people-to-3D without invented geometry

A substrate-grounded strategy, component profile, file format, verifier contract, and falsifiable test plan for chrono-spatial reconstruction from MP4 and camera images.

DESIGN PROPOSAL | NOT IMPLEMENTED | NOT PHYSICALLY VALIDATED



CORE VERDICT	A monocular video can support a partial, time-indexed, evidence-backed estimate of visible clothed surfaces. It cannot by itself verify hidden geometry, metric scale, anatomy, forces, or a complete body. The correct substrate object is an append-only evidence ledger plus sparse surfels/voxels/points and explicit UNKNOWN - not a hallucinated watertight mesh.

Repository scope: parent Git root and all relevant Grove, GPU Native, Atlas/KSEED, KLB Seedchain, and GSP4 documentation. Case study: user-supplied 9.12 s dance video, analyzed locally with classical OpenCV diagnostics; no frame images embedded.

30 AUGUST 2026 | VERSION 0.1

01 / EXECUTIVE DECISION

What is being proposed

The highest-ROI path is a refusal-capable evidence substrate first, and 3D materialization second.

Decision
Create **UGTS-VSTL-0.1** - Vision Spatial-Temporal Literal, Monocular Person Profile - as a new component profile. Its canonical state is a deterministic append-only log of accepted observations, partial visible-surface elements, uncertainty, lineage, retractions, and UNKNOWN. It is not a claim that the current repository already reconstructs dynamic people.

What "works" means

- Every committed 3D element cites independent geometric evidence, calibration, time, coordinate frame, uncertainty, and a verifier receipt.
- A failed gate leaves canonical state unchanged; learned confidence, visual plausibility, or cross-model agreement cannot self-certify geometry.
- The system may return only 2D rays, masks, tracks, and UNKNOWN when 3D is unobservable. Correct refusal is a successful result.
- Time is a sequence of exposure intervals. Spatial elements exist within slices; temporal identity is a lineage graph. No face or voxel connects different time slices.
- Metric claims require an external scale anchor. Hidden/back surfaces remain UNKNOWN unless independently observed.

Current evidence versus new work

Area	Repository evidence	VSTL decision
Static monocular mapping	Atlas has guarded relative camera motion, sparse/semi-dense points, static voxel fusion, evidence events, and KSEED export. [I3-I8]	Reuse the proposal/commit discipline; do not reuse static occupancy semantics for a moving person.
Novelty retention	UGTS-GN says novelty-only logs are valid only with authoritative state elsewhere or deterministic reconstruction. [I9-I10]	Require base/checkpoints, bounded replay depth, exact logical replay, and explicit novelty policy.
Residual replay	KLB Seedchain proves fixed-index residual/checkpoint patterns and names unstable correspondence/remeshing as kill conditions. [I11]	Reuse only where element identity is stable; otherwise checkpoint or abstain.
3D+time	Grove distinguishes chrono-spatial 3D+time from four-spatial geometry; the contract is draft and unimplemented. [I1-I2]	VSTL is explicitly 3D+time and remains a design proposal.

Case-study result
The supplied MP4 earns **VSTL-L0 OBSERVATION-EVIDENCE** for the person, with background-pose proposal diagnostics only. Person-centric relative 3D, metric 3D, and a mesh are not demonstrated.

02 / DEFINITIONS

Terms that stop category errors

The wording is deliberately narrower than common monocular human-reconstruction marketing.

Term	Normative meaning
Literal	Traceable to accepted evidence and declared constraints. It does **not** mean ground truth, complete, anatomical, photoreal, or metric.
Person object	A stable ledger entity for the **observed clothed appearance envelope**, not biometric identity and not an anatomical body.
Chrono-spatial	Three spatial dimensions indexed by measured time/exposure intervals. It is not four-spatial-dimensional geometry.
Canonical geometry	Sparse points, surfels, occupied voxels, or open same-time faces whose support passed all gates. Hidden connections are prohibited.
Derivative	A PLY, OpenVDB, glTF, OBJ, USD, render, splat, interpolated mesh, or animation generated from canonical state. It has no authority to mutate or certify the state.
Proposal	Any uncommitted mask, track, depth, pose, correspondence, body rig, mesh, or physics hypothesis from OpenCV or a learned model.
UNKNOWN	Unseen, occluded, uncalibrated, ambiguous, ill-conditioned, blurred, or otherwise unsupported. UNKNOWN is not EMPTY.
Negative memory	A new VSTL term, absent from current repository vocabulary: a time-bounded verified contradiction/retraction of prior subject occupancy. It is separate from novelty.

Canonical state

For person ledger entity P and time interval t:

O(P,t) = (E_t, L_t, N_t, U_t)

Symbol	State
E_t	Evidence-supported partial spatial elements at t: points, surfels, occupied voxels, or open same-time faces.
L_t	Verified temporal lineage edges between elements or track branches. Lineage is not geometry.
N_t	Verified time-bounded retraction/free-ray evidence with visibility proof and uncertainty.
U_t	Unobserved, occluded, ambiguous, or rejected regions. UNKNOWN remains first-class and can dominate the object.

Critical downgrade
The report uses **physically admissible, uncertainty-bounded geometric and kinematic hypotheses**, not "actual physics reconstruction." RGB video alone cannot measure mass, force, torque, muscle state, contact force, bone geometry, or material properties.

What monocular RGB can and cannot observe

The hard laws are respected by refusing unsupported claims, not by adding stronger priors.

One pixel supplies a ray, not a point

For pixel p in frame k with camera center c_k, rotation R_k, and intrinsics K_k, every positive depth lambda lies on:

r(k,p,lambda) = c_k + lambda R_k K_k^-1 p_tilde, lambda > 0

A 3D element **MUST NOT** become VERIFIED when its depth is supported only by one RGB observation or by one or more learned predictors. Verification requires independently conditioned multi-view geometry of the same material surface, measured depth, or an explicit measured constraint. Until then, the record remains a 2D/ray/silhouette constraint with UNKNOWN depth. Classical non-rigid structure-from-motion itself needs deformation and temporal assumptions and remains ambiguous when correspondence fails. [E10-E12]

Boundary	Consequence for VSTL
Projective scale gauge	Translation and shape are only relative until a known distance, calibrated multi-camera baseline, depth, or other measured anchor is accepted.
Camera and person both move	Static-SLAM residuals mix camera motion, articulation, cloth motion, foliage, reflections, and mismatches. Separate background and subject before triangulation.
Non-rigid surface	Longer baselines improve parallax but break material-point correspondence; short baselines preserve tracks but are poorly conditioned.
Hidden/back surfaces	No ray support means UNKNOWN. A silhouette constrains occupancy along/behind rays but cannot locate a unique surface.
Exposure interval	Fast motion is integrated over shutter time; rolling shutter can make each row a different time. Exact-instant geometry may need rejection.
Lossy coded video	Compression changes pixels and may use inter-frame dependencies. Preserve coded bytes, decode profile, exact PTS, and accepted-frame decoded hashes.

Hard gates versus priors

CAN BE HARD-GATED	MUST REMAIN A HYPOTHESIS
Monotone time; valid camera rotations; positive depth; calibrated ray geometry; reprojection and silhouette residuals; explicit uncertainty; SI units only after scale anchor; integrity/replay equality.	Smoothness; low rank; skeleton/body model; piecewise rigidity; finite-speed thresholds; nonpenetration/contact; cloth regularity; bone length; albedo; unseen texture; learned depth or normals.

No conservation-law theater
A motion sequence can be checked for kinematic consistency under declared tolerances. It cannot establish conservation of energy or momentum, forces, torques, material response, or human biomechanics without the missing measurements.

Reuse contracts, not conclusions

The existing modules provide an authority spine; none is a dynamic-human reconstruction implementation.

Submodule	Verified mechanism to carry	Boundary / required change
Exact Grove 3.9.2	Seed + bounded recipe + packed state; serialized meaning; bounded error/failure; named evidence. [I1]	Rendering/ECS is downstream. The four-dimensional contract is a draft and explicitly separates 3D+time from four-spatial geometry. [I2]
Atlas 4.1.1 / 3.9.4.1	Camera2/IMU ingest, FAST/BRIEF, guarded relative motion, triangulation, adjacent-keyframe semi-dense proposals, verifier, event ledger, static voxels, KSEED. [I3-I8]	Static-world assumptions fail on dancers. Current voxel fusion only adds/averages/prunes and has no person, retraction, visibility, or UNKNOWN semantics.
KSEED 4.1	Fixed header/chunks, CRC, SHA predecessor chain, checkpoints, events, compact evidence. [I6-I7]	Raw photons and people are exogenous. Raw frames are omitted by default; VSTL needs explicit source references and dynamic element support.
UGTS-GN	Verified transitions, irreducible external novelty, novelty-proportional retention, deterministic rebuild. [I9-I10]	"Negative memory" is not defined there. Novelty-only cannot replace authoritative base state or source evidence.
KLB Seedchain	Per-index prediction residuals, checkpoints, replay/error gates, and kill rules. [I11]	Assumes stable point order/correspondence. Human cloth/hair/topology often reindexes or remeshes; high novelty can make a conventional codec better.
GSP4	Hash-linked spatial event records and identity split/merge vocabulary. [I12]	A useful event precedent, not a person geometry format or proof of identity.

What the physical phone evidence actually proves

The Atlas documentation includes physical Poco X7 Pro runs and compact KSEED/PLY outputs. One 35 s run reports about 10.97 analyzed frames/s, 12,464 voxels, a 64,733-byte KSEED, unanchored scale, and quality 0.01. Another static/moved comparison also reports quality 0.01 and warns against interpreting voxel count as dependable reconstruction. This proves device execution and evidence packaging, not dynamic-human 3D accuracy. [I13-I14]

Integration status
UGTS-VSTL-0.1 is SPECIFIED HERE ONLY. No repository component presently supplies calibrated dynamic-person non-rigid reconstruction, subject-relative retractions, cross-time lineage, or the proposed VSTL file writer/reader.

Use models to propose, never to confer authority

The non-generative core can run with classical OpenCV alone; learned modules are optional and quarantined.

Launchpad	Allowed proposal role	Never canonicalize directly
OpenCV	Calibration, distortion correction, Shi-Tomasi/FAST/ORB, KLT flow, RANSAC homography/fundamental/essential models, triangulation, PnP, bundle adjustment interfaces. [E8-E9]	RANSAC support is not proof of static structure or material correspondence; assumed intrinsics are diagnostics only.
SAM 3	Person/part masks, concept-conditioned detection, instance tracking proposals. [E1]	Mask confidence cannot certify depth, identity, occlusion, or free space.
DINOv3 / ViT features	Dense feature and correspondence proposals, track recovery, part similarity. [E7]	Feature similarity is not same-material-point proof and cannot bridge occlusion by itself.
Depth Anything 3	Depth, confidence, intrinsics/extrinsics proposals and multi-frame initialization. [E5]	Monocular metric depth and Gaussian/novel-view heads are not measurements; splats are prohibited as authority.
MoGe2	Depth/point/normal/intrinsics proposals with confidence. [E6]	Predicted metric scale, normals, and masks remain proposals even if several models agree.
SAM 3D Body	Body/part/joint/ROI and pose association proposals from a parametric rig. [E3]	The full parametric or hidden body mesh cannot enter literal geometry without independent visible-surface evidence.
SAM 3D Objects	Visualization or adversarial comparison only. [E2]	It is explicitly full shape/texture generation from one image; generated PLY/Gaussian shape is prohibited from canonical VSTL.
SAM-Body4D	Temporal/identity architecture ideas and comparison only. [E4]	Its diffusion/amodal completion path conflicts with the evidence-only authority core; "training-free" does not mean non-generative or literal.

Independent support rule
Two learned models agreeing is still one evidence class: inference from the same RGB. A proposal can become VERIFIED only through independent geometry/measurement and the ordered VSTL gates. Record model name, exact weights/hash, license, inputs, coordinate frame, confidence, and uncertainty for audit.

Recommended initial stack

- FFmpeg/ffprobe for immutable coded-byte ingest, PTS/DTS/decode receipts, and deterministic frame extraction.
- OpenCV classical baseline for calibration, background/foreground motion separation, tracks, RANSAC proposals, triangulation, and reprojection checks.
- Optional SAM 3 masks and DINOv3 correspondences behind a proposal interface; disable without changing canonical semantics.
- Optional DA3/MoGe2/body-model proposals only for prioritization and diagnostics; never for hidden-surface fill or direct geometry commits.

Failure patterns the profile must make impossible

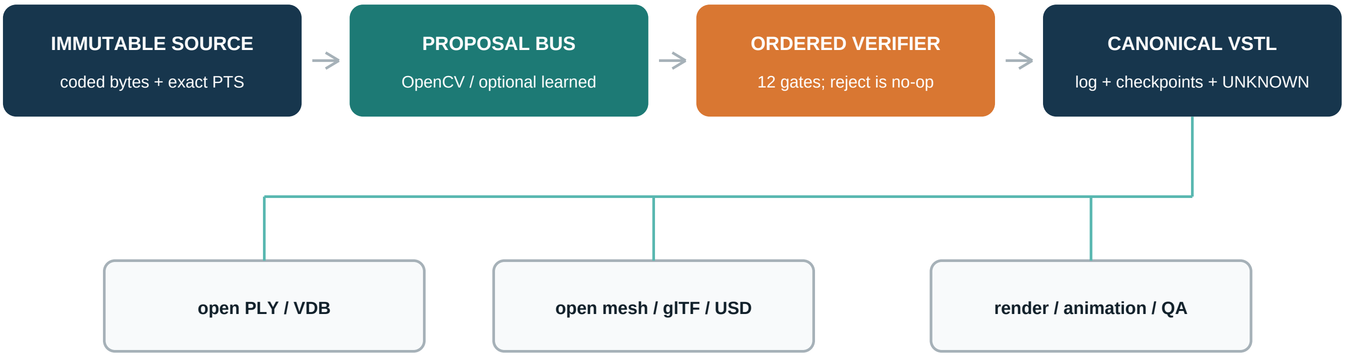
A credible design starts by naming the shortcuts that would produce an attractive but scientifically false object.

Shortcut	Why it fails	Required response
Single-image body mesh as truth	Depth and hidden shape are inferred, not observed.	Keep as proposal/visualization; commit only independently supported visible elements.
Learned depth as metric measurement	Scale and depth can be model priors with scene-dependent error.	Require measured scale and multi-view residual gates; otherwise relative/UNKNOWN.
Fuse dancers into Atlas static voxels	Person and camera motion violate static-world occupancy.	Mask/branch motion; maintain subject-relative, time-indexed state.
Intersect silhouettes across changed poses	Different time slices do not constrain one fixed visual hull unless deformation is verified.	Use only within verified motion-compensated windows; otherwise ray constraints.
Missing observation means empty	Occlusion, crop, blur, and mask failure are not negative evidence.	Map NOT_OBSERVED and OCCLUDED to UNKNOWN.
Confidence self-certifies	Model confidence is correlated with the model and input.	Use independent geometry plus verifier receipts.
Per-frame scale	The body appears to breathe and drift as gauge changes.	One accepted gauge/anchor per connected interval; branch when lost.
Bridge unknown regions with faces	A watertight sheet hides missing evidence and invents topology.	Canonical surfels/voxels/open faces only; mesh closure derivative and labeled.
Hash equals authenticity/identity	A digest proves byte consistency, not authorship, trusted time, consent, or person identity.	Use signatures/trusted time/consent records separately; IDs remain ledger-local.
Novelty-only without a base	Events cannot recreate photons or absent initial state.	Retain source refs plus bounded checkpoints; prove deterministic replay.
Smoothness equals physics	Smooth motion can still be wrong; real motion can be abrupt.	Treat smoothness and skeletons as declared priors, never laws.
Gaussian splat is the literal object	Splats can optimize appearance and novel views without evidential surface semantics.	Keep splats downstream and non-authoritative.

Kill the build, not the evidence
If correspondence is unstable, topology repeatedly reindexes, uncertainty grows without bound, or VSTL novelty/checkpoints exceed a conventional video/geometry codec at the same fidelity, preserve the source and ledger but stop claiming a reconstructive substrate win.

Propose broadly, commit narrowly

All algorithms can be replaced; the evidence-to-authority transition is the stable substrate contract.



Authority flows one way. Derivatives never certify or mutate the canonical state.

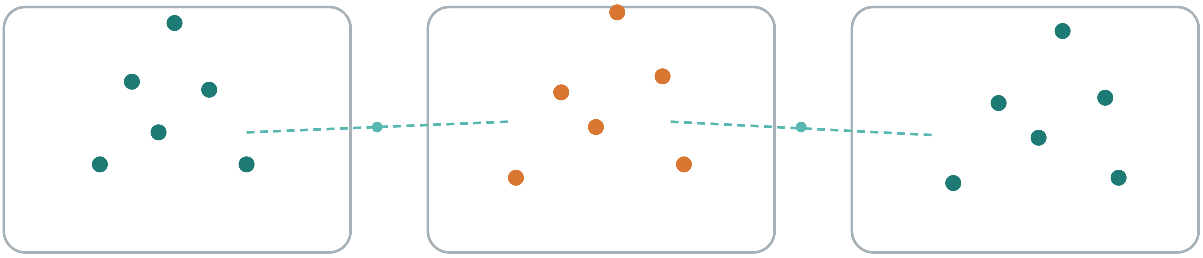
Seven bounded stages

Stage	Input -> output	Authority rule
1. Ingest	MP4/images -> coded-byte digest, stream table, exact PTS/exposure intervals, decode profile, decoded-frame receipt	Never discard the link to the coded source. Preserve edit lists/rotation/decode dependencies.
2. Calibrate	Camera metadata/targets/background -> intrinsics, distortion, rolling-shutter/exposure policy, pose proposals	Unknown calibration lowers conformance; assumed K is diagnostic only.
3. Propose	Frames -> masks, tracks, features, depths, rigs, rays, correspondences	Every module writes proposals with provenance; none mutates canonical state.
4. Separate	All motion -> background camera solution + subject/other-dynamic branches	Static pose needs sufficient independent background support; ambiguous pixels stay unassigned.
5. Verify	Proposal + prior state -> accepted event or rejection receipt	Run ordered gates; rejection is a deterministic no-op.
6. Commit/replay	Accepted event -> hash-linked VSTL state + bounded checkpoints	Pre/post hashes and exact logical replay are mandatory.
7. Materialize	Canonical slices -> PLY/VDB/open mesh/glTF/USD/render	Derivative carries source VSTL digest and coverage/non-authority label.

State mutation invariant
Only a verifier receipt may change canonical state. Proposal code, rendering code, file exporters, UI edits, and learned modules cannot bypass the ledger. Each accepted transition names its identifiers, support, compatibility, guard, confidence/error, uncertainty, effect, and pre/post state digests.

A literal entity without a connected time sheet

The object is a state-and-lineage ledger. Geometry belongs to a time slice; identity is not represented by faces stretched through time.



Dotted edges are lineage assertions. They are not geometry and do not fill occluded time or space.

Normative topology rule

- Spatial adjacency is permitted only inside one declared time interval. Same-time open triangle patches may exist when every vertex/face has support.
- A face, edge, voxel cell, tetrahedron, or splat **MUST NOT** connect samples from different times in canonical state.
- Temporal continuity is an explicit lineage edge with confidence, parent/child IDs, time gap, verifier receipt, and split/merge/reappear status.
- During occlusion or track ambiguity, branch identity or mark it UNKNOWN; never force association to preserve a pretty trajectory.
- If literally no spatial sheet is desired, surfels or occupied voxels are the canonical materialization. An open mesh remains an optional same-time derivative.

Minimum element record

Field family	Required content
Identity	element_id, entity_id, element type, parent lineage IDs, local tracker ID status (not biometric identity)
Space/time	coordinate-frame ID, exposure interval or validity interval, quantized position/normal/radius or voxel key/size
Evidence	source/stream/frame/PTS references, pixel/ray/view direction, mask/track/correspondence refs, calibration and camera-pose hashes
Uncertainty	covariance/error bound, quantization step, visibility class, scale policy, confidence separated from verifier status
Authority	proposal provenance, gate results, accepted/rejected status, pre/post state hashes, event sequence
Appearance	time/view-dependent radiance sample with color-space/decode receipt; never silently relabel as intrinsic albedo

Identity boundary
A stable VSTL entity ID means "these accepted observations belong to the same ledger branch under the stated uncertainty." It is not a person's legal identity, face identity, gait identity, or proof that an occluded reappearance is the same individual.

Two orthogonal axes

Novelty asks whether an event must be stored. Negative evidence asks what state change the evidence supports.

Repository correction
The phrase **negative memory** is not defined in the audited substrate documentation. UGTS-GN defines external novelty and novelty-proportional retention. VSTL therefore introduces negative-memory event semantics explicitly and does not pretend they already exist. [I9-I10]

Status/event	Meaning	Commit rule
NOT_OBSERVED	No accepted observation covers this element/region at t.	Always UNKNOWN. Never a removal or free-space claim.
OCCLUDED	A nearer accepted surface blocks the ray or association is ambiguous.	UNKNOWN until reobserved; preserve lineage hypothesis separately.
ADD_OCCUPIED	New visible-surface element with accepted depth and support.	Must pass calibration, same-material, conditioning, residual, uncertainty, and novelty gates.
MOVE_RESIDUAL	Accepted element differs from deterministic predictor beyond threshold.	Store quantized residual plus support; checkpoint when correspondence/gauge changes.
RETRACT_OCCUPIED	A previously accepted subject element no longer belongs to the subject during a bounded interval.	Needs calibrated visibility/occlusion proof, depth ordering, independent support, and uncertainty. It does not prove the world voxel is empty.
FREE_RAY	A calibrated subject-free ray segment over a bounded interval.	Commit only if foreground mask, occlusion, and depth limits are independently supported.
SPLIT / MERGE / REAPPEAR	Lineage changes with explicit ambiguity and evidence.	Never infer biometric identity; branch when entropy exceeds policy.
CHECKPOINT	Authoritative partial state at sequence c.	Required to bound replay and survive high novelty/reindexing.

Novelty-only replay

Let X_hat_t = Phi(X_c, t; theta) be a deterministic declared predictor from checkpoint c. Store an event when a supported state class changes, uncertainty class changes, or quantized residual exceeds policy epsilon:

delta_e(t) = x_e(t) - Phi_e(X_c,t;theta) ; X_t = Replay(X_c, N_(c+1:t))

- Replay must reproduce the identical logical state hash, event membership, and event order under declared quantization.
- Checkpoint depth is bounded by policy; source evidence remains externally referenced because no seed can recreate photons or a person.
- Quantization is conforming only when maximum error remains below the applicable guard margin.
- High novelty density is not failure of reality; it is a signal that this compression strategy has poor ROI for the interval.

10 / ORDERED VERIFIER

Twelve gates before one state transition

The exact order is part of the component profile; a rejection records why and mutates nothing.

#	Gate	Minimum acceptance question
1	Identifier and provenance	Are entity/element/model/source IDs known, hashed, licensed, and bound to the proposal?
2	Time and decode	Are coded source, stream, PTS/exposure interval, decode profile, and frame receipt valid and ordered?
3	Camera/calibration	Are intrinsics, distortion, pose frame, rolling-shutter policy, and scale mode sufficient for this claim?
4	Local support	Do source pixels/rays, masks, tracks, and observation weights directly support the proposed element?
5	Compatibility	Is the proposal compatible with person/part/element type and current lineage state?
6	Observability	Is parallax/conditioning sufficient, with same-material correspondences and positive-depth support?
7	Visibility/negative	Are occlusion, depth ordering, foreground mixing, and any retraction/free-ray claim actually verified?
8	Residual	Do reprojection, photometric, silhouette, and multiview residuals remain within bounded policy?
9	Temporal/physics	Is the transition causally ordered and physically admissible under declared - not universal - priors?
10	Uncertainty/scale	Are covariance, gauge, quantization, and metric-anchor conditions sufficient for the requested level?
11	Novelty/transition	Is the event irreducible relative to predictor/checkpoint, and is the state transition legal?
12	Capacity/integrity	Will bounded buffers, CRC/hash chain, canonical encoding, checkpoint depth, and post-state digest remain valid?

Mandatory abstention
Reject or remain UNKNOWN on decode/PTS failure; inseparable camera/person motion; insufficient static background; unknown or poor intrinsics for the requested claim; low parallax; unstable material correspondence; mask overlap/edge mixing; identity ambiguity; blur/rolling shutter beyond policy; residual failure; absent scale anchor for metric claims; replay/hash/schema failure; or materially model-dependent geometry.

Receipt fields

Every proposal receives gate-by-gate status, numeric residuals and thresholds, uncertainty, source IDs, proposal model hash, effect fields, sequence number, canonical event bytes, predecessor digest, and pre/post logical state hashes. Rejected proposals may be retained in an audit channel but never enter the authoritative replay chain as state-changing events.

Visible-surface geometry, only when earned

The baseline is classical, deterministic, and able to stop at rays or 2D tracks.

Step	Operation	Pass condition / output
A. Source bind	Hash coded MP4/images; parse streams, PTS/DTS, edit list, rotation, cadence; record decoder/version; optionally hash canonical decoded luma.	Deterministic frame table with exact exposure intervals and decode dependencies.
B. Camera model	Use metadata/calibration target; estimate distortion/intrinsics only with enough evidence; record assumed values separately.	Calibrated rays or explicit UNCALIBRATED status.
C. Motion separation	Propose target masks/tracks; estimate camera/background from static regions; model other movers as outliers/branches.	Sufficient distributed background support and bounded residual; otherwise camera pose UNKNOWN.
D. Material tracks	Track texture/surface patches with forward-back checks, appearance consistency, mask-edge exclusion, and occlusion branching.	Same-material support over a window; not merely optical-flow continuity.
E. Geometry proposal	Triangulate conditioned rays; optionally fit local non-rigid/piecewise-rigid predictors; learned depth/rigs only initialize.	Positive depth, parallax, reprojection, silhouette, and uncertainty gates.
F. Literal fusion	Commit partial surfels/voxels/points per time slice; update radiance and covariance; never fill hidden regions.	Accepted event receipt and post-state hash.
G. Novelty/replay	Predict from checkpoint; encode irreducible residual/status changes; checkpoint on gauge/topology/correspondence break.	Exact bounded replay with quantization below guard margin.
H. Derivative	Generate open point/surfel/mesh/voxel sequence for DCC, visualization, or analysis.	Carries VSTL digest, level, UNKNOWN coverage, and NON-AUTHORITATIVE label.

Capture protocol that materially improves observability

- Calibrated lens/intrinsics and known rolling-shutter/exposure behavior; preserve original timestamps and full resolution.
- A textured static background and distributed scale anchors/floor targets; avoid assuming paving, body height, or architecture dimensions.
- Slow, full-body, multi-angle coverage with repeated material patches, controlled light, low blur, limited occlusion, and a target selected before motion.
- A calibration/reference segment and, for validation, synchronized depth or multi-camera/marker truth that is never fed to the monocular inference path.
- Explicit consent from every visible person, with face exclusion and bounded retention where possible.

Visual hull caveat
Silhouette intersection is valid only for a rigid/static shape under calibrated multi-view observations. Across changing human poses it requires verified deformation compensation; otherwise each silhouette remains a time-local ray constraint. [E14]

Admissible motion, not invented mechanics

Observation is primary. Physics may reject an impossible proposal; it may not supply exact hidden geometry.

Class	Allowed normative treatment	What remains UNKNOWN
Time and causality	Monotone coded timestamps; exposure intervals; row-time model when known; lineage cannot precede its evidence.	Exact instant within an exposure when shutter/readout is unknown.
Euclidean geometry	Valid rotations, coordinate transforms, positive depth, calibrated rays, bounded reprojection error.	Absolute scale without anchor; hidden surface position.
Kinematics	Finite displacement over finite time and application-specific velocity/acceleration bounds, recorded as policies.	Forces, torques, impulses, control inputs, muscle state.
Articulation	Measured length invariants or rigid local patches may be used where evidence supports them.	Anatomical skeleton, joint center, bone length, body shape inferred from clothing.
Contact/nonpenetration	May reject obvious conflicts only when both surfaces and depth ordering are observed with uncertainty.	Contact force, friction, pressure, hidden self-contact.
Appearance/material	Time/view-dependent decoded radiance samples with color metadata.	Albedo, reflectance, fabric parameters, translucency, hair volume.
Conservation laws	No direct claim from RGB. Use only inside a separately measured simulation profile.	Mass, energy, momentum, material constitutive law.

Physics policy object

A conformance claim binds to a named policy hash containing unit mode, accepted scale anchors, camera/exposure model, velocity/acceleration limits by application, collision uncertainty, residual thresholds, quantization, and abstention behavior. Changing the policy changes the claim and invalidates silent comparison.

> **Hard-law compliance**
> The profile does not violate hard scientific laws because it never upgrades underconstrained priors into measurements. It preserves causal ordering, valid geometry, uncertainty, and scale gauge, and it marks missing physics as UNKNOWN.

Comparison point

DynamicFusion demonstrates dense non-rigid reconstruction using RGB-D, where depth is measured. It is useful as a contrast: removing depth from the sensor changes the observability problem and prevents the same authority claims. [E13]

Append-only authority core; exchange wrapper optional

The format is optimized for crash-safe evidence, deterministic replay, partial geometry, and explicit UNKNOWN - not broad interchange convenience.

Proposed identifiers
Component profile: **UGTS-VSTL-0.1.0-PROPOSAL**. Canonical stream magic: **VSTL1**. Canonical extension: **.vstl**. Optional distribution bundle: **.vstlpack**. These identifiers are new and not implemented in the repository.

Layer	Encoding target	Reason / constraint
File header	Fixed 256-byte little-endian VHDR; magic/version, feature flags, canonical schema and policy hashes, source-manifest hash, unit/gauge mode, first chunk offset, header CRC.	Stable discovery and bounded parsing. Exact byte offsets freeze only after prototype and golden-vector review.
Chunk envelope	Fixed 128-byte little-endian envelope; type, flags, sequence, time interval, stored/decoded lengths, entity/frame IDs, predecessor SHA-256, stored/decoded CRC32C, envelope CRC.	Append-only recovery, predecessor integrity, bounded allocation, independent stored/decoded checks.
Metadata payload	RFC 8949 deterministic CBOR; sorted map keys; no indefinite lengths; defined integer widths; no NaN/infinity; -0 normalized.	Avoid JSON float/canonicalization ambiguity while retaining extensibility.
Geometry payload	Fixed little-endian typed arrays. Quantized int32 positions relative to declared tile origin/step; canonical float32 covariance only under strict normalization; explicit UNKNOWN/visibility bitmap.	Deterministic bytes and bounded error. OpenVDB background values cannot silently mean FREE.
Event log	Canonical event chunks with source refs, gate receipts, pre/post state hashes, lineage, retraction/unknown semantics, and effect fields.	Only accepted state-changing events participate in authoritative replay.
Checkpoint/index	Periodic state snapshot, maximum replay stride, terminal summary/index, logical state digest and coverage statistics.	Bounded rebuild and random access without pretending a seed recreates photons.
External source	Content-addressed coded MP4/image reference; optional encrypted local object store. Source bytes need not be embedded.	Privacy and size boundary; no redundant DEFLATE of already compressed MP4.

Why ZIP/JSONL/Parquet are not the authority

ZIP64 central-directory finalization is weak for append-only crash recovery; JSON numbers are not byte-canonical without a separate scheme; Parquet bytes can vary by writer/version; PLY lacks time, uncertainty, lineage, and UNKNOWN semantics. These remain useful in the optional delivery/analysis layer, not the normative authority core.

Chunks, receipts, and delivery layout

Human-readable examples are diagnostic views; canonical bytes are deterministic CBOR and fixed arrays.

Chunk	Purpose	Required highlights
SRCM	Source manifest	Coded-byte SHA-256, size, container/codec, stream IDs, time base, frame PTS/DTS/decode dependency table, decode profile, rights/retention refs.
CALB	Calibration	Intrinsics/distortion, coordinate frames, pose gauge, scale anchors, rolling-shutter/exposure model, covariance, provenance.
OBS2	2D observation	Frame/ray/mask/track/feature refs, visible/occluded/unknown class, radiance sample, proposal provenance.
GEO3	Partial geometry	Point/surfel/voxel/open-face typed arrays, time interval, entity, quantization, uncertainty, support receipt IDs.
EVNT	Accepted transition	Event type, parents, novelty residual/status change, gate results, pre/post hashes, predecessor digest.
REJT	Non-authoritative audit	Rejected proposal and gate diagnostics. Excluded from state mutation and state hash unless policy explicitly hashes audit sidechannel.
CHKP	Checkpoint	Complete authoritative partial state, predictor/policy hash, replay root, coverage/unknown bitmap, maximum permitted stride.
INDX	Terminal index/summary	Chunk offsets, time/entity ranges, final state digest, coverage and conformance level, integrity result.

Illustrative accepted event

```
# diagnostic JSON view - canonical event is deterministic binary/CBOR
{
  "seq": 1842, "op": "RETRACT_OCCUPIED",
  "entity": "person-track:7", "element": "surfel:04ab",
  "interval_us": [7486912, 7526736],
  "support": ["frame:0188/ray:642,319", "frame:0189/ray:640,320"],
  "visibility": "VISIBLE_WITH_ACCEPTED_DEPTH_ORDER",
  "depth_basis": "MULTIVIEW_SAME_MATERIAL_POINT",
  "uncertainty_policy": "sha256:...",
  "gates": "all-pass", "pre_state": "sha256:...",
  "post_state": "sha256:...", "novelty": "status-change"
}
```

Optional delivery layout

```
case.vstlpack/                    # optional delivery, not authority
  authority/session.vstl            # canonical append-only VSTL1 stream
  sources/source-manifest.cbor       # refs; MP4 may stay in encrypted CAS
  receipts/verification-summary.cbor  # derived readable summary
  derivatives/visible-surfaces.ply     # open/partial, non-authoritative
  derivatives/animation.usd            # optional, non-authoritative
  derivatives/coverage.parquet         # analysis table, non-authoritative
  manifest.sha256                      # integrity only, not authorship/time
```

Highest ROI by role

One format should not serve evidence authority, analysis, editing, and runtime delivery at once.

Format / application	Role and ROI	Authority caveat
.vstl (new)	Highest ROI for evidence ledger, partial geometry, UNKNOWN, lineage, novelty, replay, and verifier receipts.	Canonical after implementation and conformance; currently a proposal.
MP4/images + external CAS	Highest ROI for immutable source retention and reprocessing. Keep coded bytes and exact timing; encrypt/restrict.	Source evidence, not 3D state. A hash is integrity, not consent/authenticity.
PLY + CloudCompare/MeshLab/Open3D	High ROI for inspecting partial visible points/surfels and numeric QA.	PLY normally loses time/lineage/uncertainty; open surfaces only; carry VSTL digest sidecar.
OpenVDB + Houdini	Conditional ROI for sparse occupancy/coverage workflows and effects.	Store UNKNOWN separately; default/background must never silently mean FREE.
glTF/GLB + Blender/Unity/Unreal	High ROI for derived preview, animation blocking, runtime display, and editable DCC work.	glTF favors renderable surfaces and lacks evidence semantics; label non-authoritative and do not close holes.
USD + Blender/Houdini/Maya	High ROI for time-sampled derivative scenes, variant layers, and DCC pipeline exchange.	Useful lineage metadata does not make interpolation or topology authoritative.
Parquet/Arrow + Python/R	High ROI for gate metrics, event density, coverage, uncertainty, and benchmarking.	Analytics bytes vary by writer; never the replay authority.
Gaussian splat formats	Low ROI for this literal profile; perhaps visualization comparison only.	Appearance optimization is not evidential surface reconstruction; prohibited as canonical.
KSEED / UGNL	High-value design precedents and possible adapters for shared event conventions.	Do not overload existing static/spatial formats with unimplemented dynamic-human semantics.

Recommended toolchain split

Layer	Preferred tools	Reason
Ingest + classical baseline	FFmpeg/ffprobe, OpenCV	Deterministic source/timing audit, calibration, tracks, RANSAC, triangulation, transparent failure metrics.
Optional proposals	SAM 3, DINOv3, DA3/MoGe2, SAM 3D Body	Improve proposal recall or initialization while staying replaceable and non-authoritative.
QA	CloudCompare/MeshLab/Open3D, Parquet notebooks	Visible-point, coverage, uncertainty, replay, and residual inspection.
DCC/runtime	Blender/USD; glTF/GLB to Unity/Unreal	Practical derivatives once evidence and UNKNOWN remain traceable; no feedback into canonical state.

Evidence levels, not visual quality levels

Conformance attaches to a produced artifact and its receipts, not merely to an input clip or algorithm name.

Level	Minimum artifact	Permitted claim	Prohibited leap
L0 OBSERVATI ON-EVIDENCE	Bound coded source, exact timing/decode, calibrated or uncalibrated rays, 2D masks/tracks, UNKNOWN/occlusion ledger, deterministic replay.	Timestamped 2D evidence and proposal diagnostics.	No person 3D, scale, hidden surface, or physical measurement.
L1 RAY/CAMERA	L0 plus accepted relative camera trajectory and conditioned sparse static-scene rays/points. Person geometry remains separate unless individually supported.	Relative camera/background structure in an explicit gauge.	No metric claim; no dynamic-person 3D by association.
L2 PARTIAL VISIBLE 3D	L1 plus independently verified same-material dynamic-person elements, uncertainty, lineage, and retractions per time slice.	Relative partial visible clothed-surface estimate.	No hidden/back completeness, anatomy, or watertight body.
L3 METRIC PARTIAL 3D	L2 plus accepted measured scale anchor(s), metric error bounds, and independent controlled ground-truth validation.	Metric partial visible-surface estimate within stated error/coverage.	No medical, forensic, safety, or complete-body claim without separate domain validation.
L4 DERIVED MA TERIALIZATION	Any lower level plus PLY/VDB/open mesh/glTF/USD/render carrying source digest, coverage, level, and non-authority label.	Usable visualization/animation/analysis derivative.	Visual completeness cannot upgrade evidence level.

Coverage is part of the claim

Each artifact reports visible-supported fraction, occluded fraction, not-observed fraction, ambiguous/rejected fraction, scale/gauge mode, time coverage, and element uncertainty distribution. A system that returns fewer but well-supported elements may outrank a dense plausible mesh for literal use.

No "full 3D" shorthand
No conformance level permits an unqualified complete-body claim. L2/L3 are explicitly partial visible-surface estimates. L4 is a derivative level, not an evidence upgrade.

Case-study classification

Artifact	Result for case-study-video-01
L0	ACHIEVABLE from this audit: coded-byte hash, exact 229-frame PTS table, 2D motion/quality diagnostics, explicit UNKNOWN/occlusion policy.
L1	BACKGROUND-POSE CANDIDATE only. Requires real calibration, explicit foreground exclusion, verified SfM/BA, and error bounds. Not earned here.
L2	NOT DEMONSTRATED for the person. Track/parallax tradeoff and non-rigid motion prevent verified same-material dynamic geometry.
L3	REJECTED for this run: no scale anchor, intrinsics, depth, IMU, or independent 3D truth.
L4	NOT PRODUCED. A mesh would be visually plausible at best and would not raise conformance.

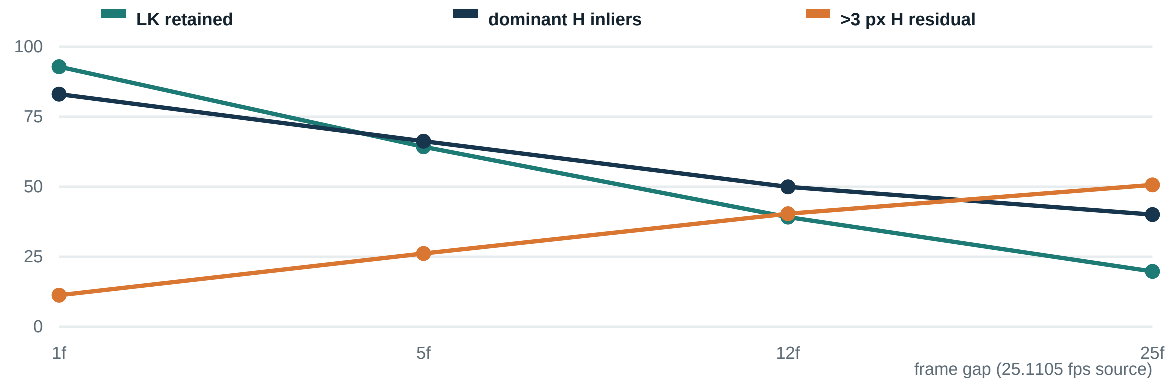
case-study-video-01: failure-aware classical probe

No learned or generative models were used. No face/frame imagery is embedded in this report.

Property	Measured value
Source digest	SHA-256 1867BAFA7C80C31F18856525CBF580EDAA36D524270B1FA59CC643B51964CBFD
Container/stream	MP4/QuickTime; one H.264 Main progressive yuv420p 8-bit video stream; no audio
Raster/rate	1280 x 720; 229 decodable frames; 9.119696 s; exact 0.039824 s PTS step; 25.110486 fps
Coding	One I-frame at t=0 followed by 228 P-frames; approximately 10.62 Mbit/s
Missing metadata	No intrinsics, distortion, exposure, shutter/readout, IMU, GPS, scale, rotation, timecode, or trustworthy capture-time tags
Observed content	One continuous outdoor dance shot; moving camera; several independently moving people; loose clothing/hair; partial crops; self/other occlusion; changing scale; hard shadows; motion blur; no rear-surface coverage

OpenCV 4.13 methodology

All 229 frames were decoded and downsampled to 640 x 360. Shi-Tomasi features (max 2,200, quality 0.008, minimum distance 6 px) were tracked with pyramidal Lucas-Kanade (21 x 21, 3 levels) and a forward-back gate under 1.5 px. Dominant homographies used 2 px RANSAC; residual outliers exceeded 3 px. Diagnostic essential matrices assumed f=512 px, centered principal point, zero distortion and therefore have no calibration authority. Pair counts were 228 at 1 frame, 45 at 5, 19 at 12, and 9 at 25.



Gap	LK retained	H inliers	>3 px H residual	Indicative ray angle	Positive-depth support
1f / 0.040 s	92.9%	83.1%	11.3%	0.453 deg	2.4%
5f / 0.199 s	64.3%	66.3%	26.2%	0.631 deg	24.4%
12f / 0.478 s	39.2%	50.0%	40.4%	0.800 deg	53.9%
25f / 0.996 s	19.8%	40.1%	50.7%	1.408 deg	54.8%

The longer baseline raises nominal ray angle and cheirality while destroying track survival and dominant-rigid support. This is the central monocular non-rigid tradeoff, not a reconstruction success metric.

What the supplied clip proves

The value of this fixture is that it pressures the system to keep ambiguity and reject seductive geometry.

Diagnostic	Measured result	Interpretation boundary
Track lifetime	Median 6 frames; 11.1% survived at least 50 frames; 5.24% survived at least 100 (14,256 tracks observed).	Optical-flow survival is not same-cloth-point identity and does not prove 3D.
Structured motion	Adjacent central H-outliers 24.7% versus 8.7% border; a second H explained median 46.0% of first-H outliers.	Consistent with independent movers, but not a certified person mask.
Background proxy	ORB/affine background fit succeeded for 228 pairs; median 1,795 matches, 75.16% inliers, 0.854 px fit residual; median 5.643 px/frame translation proxy.	Image-plane camera-motion evidence only; no calibrated 3D pose.
Subject-motion proxy	Coarse orange-clothing residual median 5.258 px, p95 35.19 px; 64.5% over 2 px.	Separates independent motion statistically; not a semantic mask or geometry measurement.
Shot continuity	Maximum adjacent HSV histogram distance 0.099; no likely cuts.	A continuous shot helps timing but not observability.
Image quality	Global Laplacian-variance proxy min/median/max 20.32/56.28/180.32; little hard clipping.	Content-dependent; fast limbs/hair still show motion blur. Rolling shutter remains UNKNOWN.

Five correct test outcomes

- Byte-identical ingest and deterministic 229-frame PTS/decode table, including the one-I-frame dependency boundary.
- Multi-person mask/track proposals that retain ambiguity, crop, occlusion, split, and reappearance rather than forcing one body track.
- Background-only relative camera-pose proposal with foreground exclusion; calibration and bundle-adjustment gates still required.
- Most person 3D proposals remain UNKNOWN or rejected because calibration, scale, same-material multi-view support, and truth are absent.
- Ledger replay reaches identical logical state and applies privacy retention/deletion policy. No attractive mesh is needed for a pass.

Case-study verdict
VSTL-L0 OBSERVATION-EVIDENCE only, with BACKGROUND-POSE-PROPOSAL diagnostics. L1 for the person is unearned. L2 relative partial visible geometry, L3 metric geometry, and L4 mesh materialization are not demonstrated.

Privacy note

The clip contains identifiable faces, bodies, and gait of multiple people. Use requires rights/consent for every visible person, local/encrypted handling, minimum retention, face exclusion where possible, and deletion controls. A hash is not anonymity; derived tracks and geometry remain sensitive or biometric-adjacent.

How to earn L2 and L3

The monocular path must be evaluated against independent truth that is never used as reconstruction input.

Test family	Required design	Primary metrics
Deterministic ingest/replay	Golden MP4/image fixtures with VFR/CFR, edits, rotation, corruption, P/B dependencies, dropped frames, and decoder versions.	PTS equality, decoded-luma receipt, event membership/order, final logical hash, checkpoint latency, corruption localization.
Controlled rigid baseline	Calibrated static object/background, known scale, monocular input; independent surveyed/depth truth.	Pose error, reprojection, scale drift, visible-surface error, false occupied/free, UNKNOWN coverage.
Controlled deforming surface	Textured cloth/mannequin/human with slow known motion; synchronized multi-camera/depth/markers as evaluation truth only.	Visible-surface 3D error, temporal drift, same-material track precision, false retraction, lineage switches.
Occlusion/negative evidence	Known occluders, crop/re-entry, self-occlusion, multiple similar people, foreground crossings.	UNKNOWN correctness, false EMPTY/free-ray rate, retraction precision/recall, split/merge/reappear accuracy.
Physics/scale	Measured scale anchors and timing; separate policies for dance, sports, ergonomics, etc.	Metric error, gauge resets, velocity/acceleration residual, policy-bound violations; no force claims.
Model ablation	Classical baseline; each optional learned proposer individually; equal verifier thresholds and evidence.	Proposal recall, accepted precision, model-dependence rate, abstention calibration, runtime/energy/license footprint.
Compression	Compare VSTL events/checkpoints against source video + conventional geometry codec at matched visible-surface error.	Bytes, novelty density, checkpoint frequency, replay time, maximum quantization error, conventional-codec crossover.

Release kill gates

- Any unsupported hidden surface enters canonical state, or UNKNOWN is collapsed into FREE/occupied without proof.
- Metric output changes materially with learned model choice despite identical observations and verifier policy.
- False retractions or forced identity links exceed application tolerance, especially under overlap and reappearance.
- Quantization changes event membership/order or exceeds a guard margin; replay state digest differs.
- Novelty/checkpoint density, runtime, or storage is worse than a conventional alternative at matched accuracy without compensating audit value.
- The application needs anatomy, forces, forensics, safety, or medical validity that the sensor/evidence cannot supply.

> **Ground truth rule**
> Multi-camera, RGB-D, motion-capture markers, surveyed targets, or calibrated turntables may be used for evaluation even though the operational input remains monocular. Without independent truth, a reconstruction can be internally consistent and still wrong.

ROI now, later, and never from this evidence

The current high-return products are auditability and partial motion evidence, not automatic complete avatars.

Readiness	Application	Why / condition
NOW - HIGH ROI	Ingest/quality/coverage assistant	Flag timing, blur, crop, occlusion, parallax, static-background support, scale-anchor absence, and likely reconstruction refusal before expensive processing.
NOW - HIGH ROI	Deterministic 2D person evidence ledger	Timestamped masks/tracks/rays, split/merge/UNKNOWN, privacy controls, reproducible model/proposal comparison.
NOW - HIGH ROI	Animation and performance reference	Use 2D/partial motion evidence and uncertainty to assist an artist; final rig/mesh is an editable derivative, not literal truth.
NOW - MEDIUM ROI	Research, teaching, algorithm QA	Excellent for studying static/dynamic separation, refusal, novelty density, retractions, replay, and model ablation.
CONDITIONAL	Controlled partial volumetric/performance capture	Requires calibrated capture protocol, scale anchors, sufficient coverage, independent truth, and demonstrated L2/L3 error bounds.
CONDITIONAL	Non-safety sports/ergonomics motion envelope	Only with consent, known scale, coverage/uncertainty thresholds, and explicit ban on anatomy/force/medical interpretations.
CONDITIONAL	Telepresence/VR visible-surface proxy	Latency and coverage can be useful if holes/UNKNOWN remain visible and no hidden completion is presented as evidence.
NOT ACCEPTABLE	Medical/rehab decisions, forensics/legal reconstruction, biometric/gait identity, safety certification	The required anatomy, causality, identity, or error guarantees are not observable or validated.
NOT ACCEPTABLE	Arbitrary-video body measurements, garment fit, unseen-avatar completion, deepfake/photoreal identity rendering	Metric scale, hidden anatomy/surface, and consent/identity risks exceed the evidence boundary.

Substrate evidence so far

The repository's practical proof is strongest in bounded proposal/commit ledgers, deterministic compact event formats, static-scene mobile capture, residual/checkpoint replay patterns, and downstream GPU rendering. Dynamic-person literal 3D remains a research program. The immediate ROI is to combine those proven mechanisms into a refusal-capable evidence format and test harness before committing to a scanner product.

Product principle
Sell or use the **coverage and certainty you can demonstrate**. Do not sell the density or attractiveness of a derivative mesh as reconstruction accuracy.

Build evidence infrastructure before reconstruction density

Each phase produces a useful artifact and has a stop condition. There is no requirement to reach L4.

Phase	Deliverable	Exit / kill condition
0. Freeze profile	VSTL terminology, source/time/calibration schema, deterministic CBOR rules, event/state semantics, 12 gates, privacy/rights policy, golden byte vectors.	Two independent readers produce identical logical hashes; malformed lengths/NaN/unknown enums fail closed.
1. L0 engine	FFmpeg ingest, exact PTS/decode receipts, OpenCV 2D masks/tracks/features, UNKNOWN/occlusion/identity branches, append-only writer/reader/checkpoints/replay.	Case-study-video-01 passes exact ingest/replay and refuses person 3D; no hidden fill.
2. Background L1	Calibrated camera profile, foreground masking, distributed static tracks, SfM/BA proposal, uncertainty and scale-gauge state.	Controlled static fixtures meet pose/reprojection bounds; supplied clip remains only a background candidate if calibration is absent.
3. Person L2 research	Same-material track verifier, local non-rigid triangulation, partial surfel/voxel slices, retraction/lineage events, controlled ground truth.	Kill windows with unstable correspondence, high false retraction, or model-dependent accepted geometry.
4. Metric L3	Measured anchors, SI-unit policy, controlled capture application profile, independent truth and coverage/error certification.	No arbitrary-MP4 metric fallback; abort when anchor/gauge is lost.
5. Derivatives L4	Open PLY/VDB, USD/glTF/GLB exporters, coverage overlays, DCC/runtime integration, source digest and non-authority labels.	Exporter cannot mutate canonical state or close UNKNOWN without an explicit visual-only flag.
6. Optimize	GPU proposal kernels, packed spatial tiles, novelty residual compression, checkpoint scheduling, device budgets.	Only after correctness; kill if conventional codecs/tools win at matched fidelity and audit value.

First executable slice

- A command-line tool that ingests MP4/images, writes SRCM/OBS2/REJT/EVNT/CHKP/INDX chunks, and reproduces the final logical state hash.
- An OpenCV background-motion diagnostic and 2D person proposal adapter; no learned dependency required for conformance.
- A verifier dashboard showing why geometry is UNKNOWN/rejected, with time, parallax, calibration, blur, occlusion, identity, and scale gates.
- A PLY exporter that emits only accepted partial visible elements and a coverage/uncertainty sidecar; no hole filling or unseen texture.
- Golden cases including the supplied dance clip as an L0 refusal fixture and controlled calibrated sequences for L1/L2 research.

Do not hide incompleteness
Atlas, KSEED, KLB, GSP4, and Grove should be imported as named precedents/adapters. The VSTL writer, dynamic-person verifier, lineage/retraction semantics, controlled ground truth, and 3D person solver are new work and must remain visibly incomplete until implemented and measured.

Literal evidence is sensitive even when it is incomplete

Geometry and tracks can remain identifying after source frames are removed.

Control	Minimum policy
Consent and rights	Explicit purpose, rights/consent for every visible person, bystander handling, permitted outputs, retention period, revocation/deletion path.
Data minimization	External encrypted source CAS; face exclusion/redaction where compatible with purpose; store only accepted evidence and required audit; bound rejected-proposal retention.
Separation	Keep person ledger IDs local and random. Do not bind to names, face recognition, gait identity, or external profiles.
Integrity	SHA-256 predecessor chain, CRC32C, canonical bytes, pre/post state hashes, reader bounds. Hashes do not provide authorship, consent, trusted time, or anonymity.
Authenticity	Optional signatures, trusted timestamp, device attestation, and consent receipts are separate layers with revocation/key policy.
Model governance	Record exact code/weights/hash/license, input refs, confidence and version; validate licenses for SAM/DINO-derived components; support a no-learned-model profile.
Access and deletion	Least privilege, encrypted transport/storage, audit logs, per-source/per-entity deletion tombstones plus rebuild, derivative inventory, and backup expiry.
Abuse boundary	No biometric/gait surveillance, identity inference, non-consensual avatar/deepfake, medical/forensic assertion, or hidden-body completion.

Deletion and replay tension

Append-only integrity conflicts with erasure. The recommended design stores source/person payloads under per-entity encryption keys and uses cryptographic erasure plus a rebuildable redacted VSTL chain whose deletion event references, but does not retain, removed content. The governance profile must state whether historical digests remain and what they reveal. This is a policy design problem, not solved by a hash chain alone.

> **Release posture**
> Until the implementation, ground-truth trials, privacy review, and license review exist, UGTS-VSTL must be distributed only as a design proposal. Any demo should show UNKNOWN, rejection reasons, coverage, and non-authority labels as prominently as visible geometry.

Repository anchors

Internal citations refer to the audited parent Git root. Paths are shown relative to that root where practical.

[I1] UGTS_KC_K_Kij_T_Grove_3_9_2_COMPLETE_ALL_FILES/docs/UGTS_SUBSTRATE_MECHANISM_MAP.md - exact retained chain, serialized meaning, bounded error/failure, and evidence requirements.

[I2] UGTS_KC_K_Kij_T_Grove_3_9_2_COMPLETE_ALL_FILES/spec/FOUR_D_CONTRACT_DRAFT.md - distinguishes chrono-spatial 3D+time from four-spatial-dimensional geometry and states the draft status.

[I3] UGTS_KC_4_1_1_ATLAS_SPATIAL_SEED_SLAM_POCO_X7_PRO_12GB_MERGED_REPAIRED_ANDROID/UGTS_AtlasSeedSLAM_411/docs/ACCURACY_AND_LIMITS.md - monocular/metric limits, verifier gates, moving-object/rolling-shutter failures.

[I4] .../UGTS_AtlasSeedSLAM_411/docs/ARCHITECTURE.md - Camera -> luma -> FAST/BRIEF -> proposals -> verifier -> map -> ledger -> KSEED.

[I5] .../UGTS_AtlasSeedSLAM_411/docs/COMPRESSION_AND_FORMAT.md - seed plus measured evidence deltas; not seed-only reconstruction.

[I6] UGTS_KC_4_1_Poco_X7_Pro_Seed_Native_Package/UGTS_KC_4_1_Poco_X7_Pro_Seed_Native/spec/KSEED_FORMAT_4_1.md, plus . ../UGTS_AtlasSeedSLAM_411/spec/KSEED_4_1_RECONSTRUCTED_FORMAT.md - fixed header/chunks, frames/events/checkpoints, CRC and predecessor hash chain.

[I7] UGTS_KC_4_1_Poco_X7_Pro_Seed_Native_Package/UGTS_KC_4_1_Poco_X7_Pro_Seed_Native/docs/SEED_STORAGE_BOUNDARY.md - seed cannot recreate photons/person/depth; raw frames off by default; hashes provide integrity only.

[I8] .../UGTS_AtlasSeedSLAM_411/provenance/SUBSTRATE_ALIGNMENT.md - proposal/commit authority lineage.

[I9] Unified_Geometric_Topological_Substrate_GPU_Native_Package/UGTS_GPU_Native_Addendum_Package/spec/UGTS_GN_1.1.md, especially lines 233-248 - novelty log valid only with authoritative or reconstructible state.

[I10] .../UGTS_GPU_Native_Addendum_Package/spec/knowledge_catalog.json - M092 external novelty; M114 novelty-proportional retention; M115 rebuild; M140 source motif 'Keep novelty only'.

[I11] klb_seedchain_gpu_v0.5.0_full_sgp4/klb_seedchain_gpu_v0.5.0_full_sgp4/docs/SEEDCHAIN_DEPLOYMENT.md - indexed residuals/checkpoints, replay/error gates, and unstable-correspondence/remeshing kill conditions.

[I12] gsp4_spatial_kd_v0.5.0/gsp4_spatial_kd_v0.5.0/docs/FILE_FORMATS.md - hash-linked spatial event records and identity split/merge event vocabulary.

[I13] .../UGTS_AtlasSeedSLAM_411/docs/POCO_X7_PRO_DEVICE_PERFORMANCE_ELI5.md - physical 35 s device run, analyzed fps, voxel/KSEED size, unanchored quality caveat.

[I14] UGTS_KC_3_9_4_1_ATLAS_POCKET_SLAM_POCO_X7_PRO_12GB_ANDROID_REPAIRED/UGTS_AtlasSLAM_3941/docs/POCO_X7_PRO_PERFORMANCE_ELI5_2026-08-28.md - physical static/moved run comparison and low-quality warning.

> **Interpretation**
> Repository citations demonstrate mechanisms and boundaries only. They do not establish an integrated VSTL implementation or human-reconstruction accuracy.

Official launchpad documentation and primary reconstruction literature

External links were checked for this strategy on 30 August 2026. Model licenses and versions must be rechecked at implementation time.

[E1] Meta AI, SAM 3: segmentation and tracking with concepts.
https://ai.meta.com/research/sam3/

[E2] Meta, SAM 3D Objects official repository: single-image full 3D shape/texture/layout generation and Gaussian PLY output.
https://github.com/facebookresearch/sam-3d-objects

[E3] Meta, SAM 3D Body official repository: single-image full-body parametric mesh recovery and optional SAM 3/MoGe2 components.
https://github.com/facebookresearch/sam-3d-body

[E4] Gao et al., SAM-Body4D project and paper: training-free 4D human reconstruction with SAM 3, SAM 3D Body, and diffusion-based amodal completion.
https://github.com/gaomingqi/sam-body4d

[E5] ByteDance Seed, Depth Anything 3 official repository: depth/confidence/camera outputs, video streaming, optional Gaussian head.
https://github.com/ByteDance-Seed/depth-anything-3

[E6] Microsoft Research, MoGe official repository: monocular geometry, depth, point maps, intrinsics, masks, normals and confidence.
https://github.com/microsoft/MoGe

[E7] Meta AI, DINOv3 official research page: self-supervised visual features for dense tasks and correspondence.
https://ai.meta.com/research/dinov3/

[E8] OpenCV official calib3d documentation: camera calibration, multi-view geometry, pose and triangulation functions.
https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html

[E9] OpenCV official optical-flow tutorial/documentation: Lucas-Kanade and dense optical flow.
https://github.com/opencv/opencv/blob/4.x/doc/tutorials/others/optical_flow.markdown

[E10] Bregler, Hertzmann, Biermann, Recovering Non-Rigid 3D Shape from Image Streams, CVPR 2000.
https://mrl.cs.nyu.edu/publications/recovering-nonrigid/bhb-cvpr00.pdf

[E11] Garg, Roussos, Agapito, Dense Variational Reconstruction of Non-Rigid Surfaces from Monocular Video, CVPR 2013.
https://openaccess.thecvf.com/content_cvpr_2013/html/Garg_Dense_Variational_Reconstruction_2013_CVPR_paper.html

[E12] Shi et al., Non-Rigid Structure-from-Motion: Temporally-Smooth Procrustean Alignment and Spatially-Variant Deformation Modeling, CVPR 2024.
https://openaccess.thecvf.com/content/CVPR2024/html/Shi_Non-Rigid_Structure-from-Motion_Temporally-Smooth_Procrustean_Alignment_and_Spatially-Variant_Deformation_Modeling_CVPR_2024_paper.html

[E13] Newcombe et al., DynamicFusion: Reconstruction and Tracking of Non-Rigid Scenes in Real-Time, CVPR 2015 (RGB-D comparison).
https://openaccess.thecvf.com/content_cvpr_2015/html/Newcombe_DynamicFusion_Reconstruction_and_2015_CVPR_paper.html

[E14] Laurentini, The Visual Hull Concept for Silhouette-Based Image Understanding, IEEE TPAMI 1994.
https://doi.org/10.1109/34.273735

FINAL POSITION

Build the ledger that knows when it does not know.

A useful substrate version of monocular people-to-3D is not a complete avatar generator. It is an evidence-bound, chrono-spatial partial object whose visible elements, uncertainty, lineage, negative/retraction events, and UNKNOWN regions can be replayed exactly. The proposed VSTL1 format and verifier make that claim testable. The supplied video is already valuable because it demonstrates the conditions under which the system must remain at L0 and refuse a person mesh.

NEXT CLAIM TO EARN: IMPLEMENTED L0 WRITER/READER + EXACT REPLAY + CONTROLLED L1 CAMERA TRUTH